

Assembly instructions

Here is the set of instructions and the allowed operands. An entry *reg* indicates that it can be any general-purpose register (eax, ebx, ecx, edx), *mem* indicates a memory location on the stack addressed as *off(%ebp)*, where *off* is the offset, and *const* indicates a constant value (constant number 4 is represented as \$4). The following tables are a quick guide to instructions and allowed operands.

movl (Moving values between registers or between register and memory or putting a constant value into a register or memory location)

movl	reg	reg
movl	mem	reg
movl	reg	mem
movl	const	reg
movl	const	mem

Arithmetic operations (addl, imull, subl: addl is for addition, subl is for subtracting the first operand from the second, and imull is an integer multiplication. The result of the operation is stored in the second operand)

addl/subl/imull	reg/const/mem	reg
addl/subl/imull	reg/const/mem	reg
addl/subl	reg/const	mem

Stack operations (pushl/popl)

pushl	mem/const/reg
popl	mem/reg

Compare instruction (compare the operands and sets the flags that can be used by the jump instructions)

cmpl	reg/mem/const	reg
cmpl	reg/const	mem

Jump instructions (jmp/jne/je/jg/jge/jl/jle)

any jump instruction	label of a basic block
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